# Simplicity as a Feature: Robust Transition-State Search by Gentlest Ascent Dynamics with Cheap Analytic Hessians

June 25, 2026


## Abstract

Machine-learned interatomic potentials (MLIPs) make analytic Hessians cheap ($\approx 0.06$ s/structure with HIP [9]), so the bottleneck in transition-state (TS) search is no longer the Hessian but the *step*. We use the minimal choice — first-order Gentlest Ascent Dynamics (GAD) on the lowest eigenvector of the Eckart-projected analytic Hessian, a Cartesian step with no trust region, quasi-Newton update, RFO, or eigenvector prediction. Our primary metric is the IRC-validated *intended-reaction success rate* (forward and backward IRC must graph-match the intended reactant and product), on 287 Transition1x test reactions from perturbed-saddle guesses (10–200 pm).

**Highlights**

- **Crossover, not a point.** GAD is statistically indistinguishable from the strongest Sella baseline when the guess is good (McNemar $p \approx 1.0$, $\leq 50$ pm) and pulls significantly ahead as it degrades ($p < 10^{-4}$ by 150–200 pm; **+21.6 pp intended at 200 pm**, 45.3% vs. 23.7%). The same crossover holds for saddle-convergence and across $F_{\max}$ thresholds.
- **Mechanism (corrected).** *No* method converges to an unintended (wrong-basin) saddle — unintended $\approx 0/287$ for all three. GAD's edge is recovering the full two-sided reaction more often; Sella's high-noise loss is non-convergence + one-sided (partial) saddles, not wrong basins.
- **Honest cost.** First-order GAD cannot tighten $F_{\max}$ below $\approx 0.01$ eV/Å; no step budget removes it. This plateau lives on the *secondary* (convergence) axis only.
- **Hybrid.** A GAD→Newton switch removes the plateau, wins wall-time per converged TS (1.4–3×) and geometry (tightest p95 RMSD), and inherits GAD's chemistry.
- **Scope boundary.** Robustness is demonstrated for *perturbed-saddle* guesses. From a reactant start the GAD family is weak (hybrid ∼2%) and Sella wins — GAD needs to start near the saddle for the reaction mode to be identifiable.


## 1 Introduction

- TSs are index-1 saddle points; locating them needs curvature (the Hessian, or at least its leftmost eigenpair). *Ab initio* Hessians are expensive, so optimizers avoid them: dimer [2], eigenvector-following / P-RFO [3, 4, 5], Sella's iterative partial-Hessian diagonalization [6, 7], and recent ML prediction of the Hessian [10, 9] or its leftmost eigenvector [11].
- **The premise has changed.** With MLIPs the analytic Hessian is cheap ($\approx$0.06 s for HIP [9]). The bottleneck becomes the **step**. Trust-region/Newton jumps ($\mathcal{O}(0.1$ Å$)$) are optimal near a quadratic saddle but fragile from a poor guess.

- **This work.** Given a cheap exact Hessian, discard P-RFO, trust radii, QN, and eigenvector prediction; step with first-order GAD [1]. Score success by IRC-validated *intended* recovery [10, 11], not mere convergence. Thesis: *in the MLIP regime you need not avoid the Hessian — you need to step more carefully with it.*

## 2 Methods

**GAD dynamics.** With forces $\mathbf{F} = -\nabla E$ and $\mathbf{v}_1$ the lowest eigenvector of the mass-weighted, Eckart-projected Hessian,

$$\dot{\mathbf{x}} = \big(\mathbf{I} - 2\,\mathbf{v}_1\mathbf{v}_1^\top\big)\mathbf{F} = \mathbf{F} - 2(\mathbf{F}\cdot\mathbf{v}_1)\,\mathbf{v}_1, \qquad \mathbf{x}_{k+1} = \mathbf{x}_k + dt\,\dot{\mathbf{x}}. \tag{1}$$

- **Step:** fixed-$dt$ explicit Euler ($dt \in \{0.003, 0.005, 0.007\}$; representative 0.005), Cartesian, no history/trust-region/line-search; mode tracked by max overlap; per-atom cap 0.35 Å.
- **Potential:** HIP [9], an $SE(3)$-equivariant MLIP returning the analytic Hessian ($\approx 0.06$ s); not trained on Transition1x (held out); Hessian recomputed every step.
- **Eckart projection:** remove the 6 translation/rotation modes in mass-weighted coordinates, diagonalize the reduced ($3N-6$) Hessian; single most important robustness ingredient. (Cartesian vs. MW step: $\leq 0.35$ pp.)
- **Hybrid GAD–Newton:** walk with Eq. (1); near a clean index-1 saddle ($n_{\text{neg}}$=1, small force) switch to a trust-radius-capped eigenvector-following Newton step on the same analytic Hessian.
- **Benchmark:** 287 Transition1x [8] test reactions; start = true saddle + isotropic Gaussian noise $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm; baseline = Sella [7] (strongest config: Cartesian + Eckart, exact Hessian/step; Appendix D).

**Two metrics, kept deliberately separate.**

- **Intended-reaction success rate (IRC) — primary.** Did we find the *right* TS? After optimization, follow Sella's IRC [7] forward/backward and graph-match each endpoint (Open Babel connectivity + VF2) to the labelled reactant/product. Outcomes (after [10, 11]): *intended* (both endpoints match), *partially intended* (one), *unintended* (valid saddle, other reaction), *no reaction* (collapsed to a minimum), *TS error* (no convergence). This is the chemically meaningful score and our headline metric.
- **Saddle-convergence rate — secondary.** Did the optimizer reach *a* first-order saddle at all: $n_{\text{neg}}$=1 $\wedge$ $F_{\max}$<$\tau$ (default $\tau = 0.01$ eV/Å; threshold family in Appendix C). A pure-geometry check, necessary but *insufficient* for the intended reaction; reported separately in Section 3.3, not alongside the primary metric.

# 3 Results

## 3.1 Primary result: intended-reaction success (IRC) crosses over with guess quality

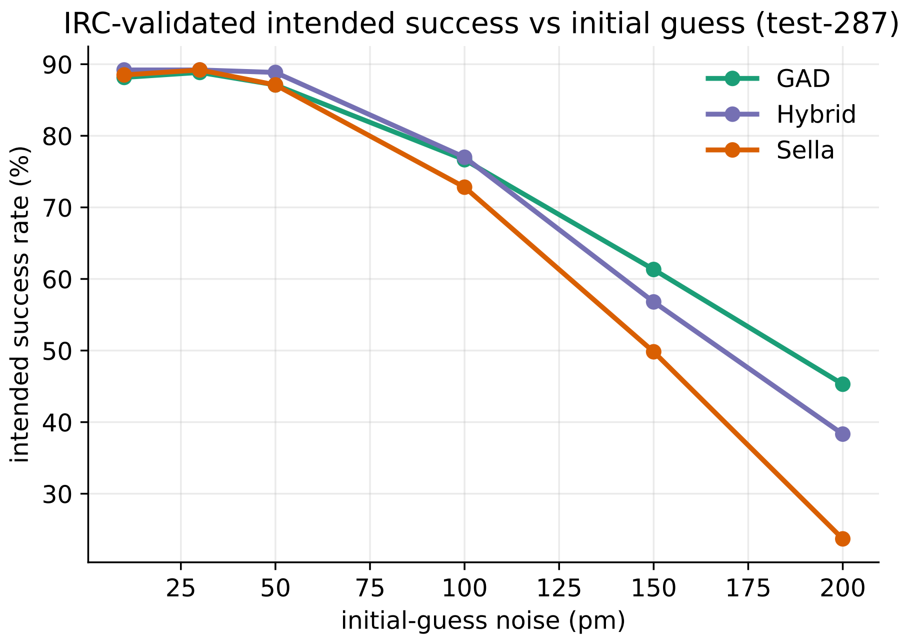


**Figure 1: Primary metric: intended-reaction success rate (IRC) vs. initial-guess noise**, $n = 287$. Methods coincide at low noise; GAD and the hybrid retain far more intended reactions as the guess degrades. (Saddle-convergence is shown separately in Fig. 7.)

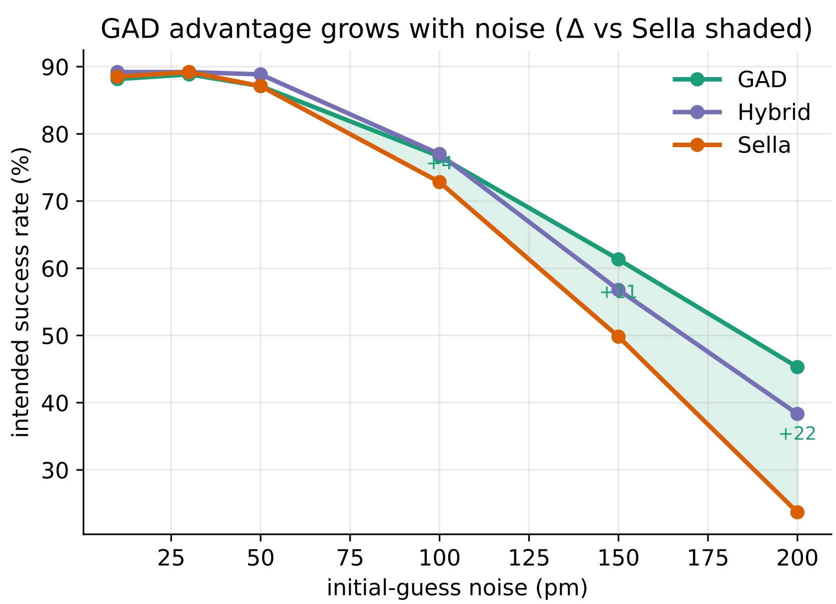


**Figure 2: Intended success for all three methods; the GAD−Sella gap is shaded.** Paired-McNemar (GAD vs. Sella): indistinguishable $\leq 50$ pm ($p$=1.0), emerging at 100 pm ($p$=0.10), highly significant at 150 pm ($p$=$3.8\times10^{-5}$) and 200 pm ($p$=$1.7\times10^{-13}$).

- **Crossover.** Tie at low noise (intended 88–89% at 10 pm); GAD pulls ahead with degradation: $\Delta$intended $= +3.9, +11.5, +21.6$ pp at 100/150/200 pm (Fig. 2, Table 1).
- **Significance.** McNemar on paired per-sample intended outcomes: n.s. at $\leq 50$ pm ($p$=1.0), emerging at 100 pm ($p$=0.10), highly significant at 150 pm ($p$=$3.8 \times 10^{-5}$) and 200 pm ($p$=$1.7 \times 10^{-13}$). The same crossover appears on saddle-convergence and across $F_{\max}$ thresholds (Section 3.3).

**Table 1: Primary metric — intended-reaction success rate (IRC), %**, $n = 287$, from the fresh per-sample re-validation (GAD $dt$=0.003); Sella = strongest baseline. Saddle-convergence is reported separately in Table 2 (Section 3.3).

| **Intended success %** | **10 pm** | **30 pm** | **50 pm** | **100 pm** | **150 pm** | **200 pm** |
|---|---|---|---|---|---|---|
| GAD (ours) | 88.2 | 88.9 | 87.1 | 76.7 | **61.3** | **45.3** |
| Sella (baseline) | 88.5 | 89.2 | 87.1 | 72.8 | 49.8 | 23.7 |
| $\Delta$ (pp) | −0.3 | −0.3 | 0.0 | **+3.9** | **+11.5** | **+21.6** |

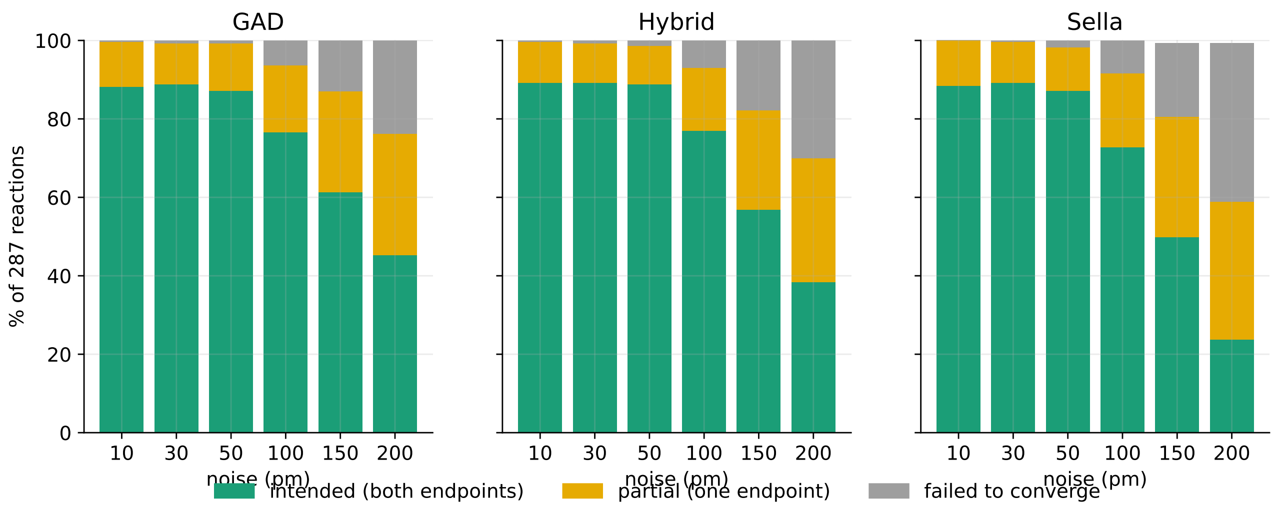


**Figure 3: Full IRC outcome breakdown.** *Unintended* $\approx 0$ for every method at every noise, so the third band is purely non-convergence (TS error / no reaction). Sella both fails to converge *and* lands one-sided (partial) saddles more often at high noise.

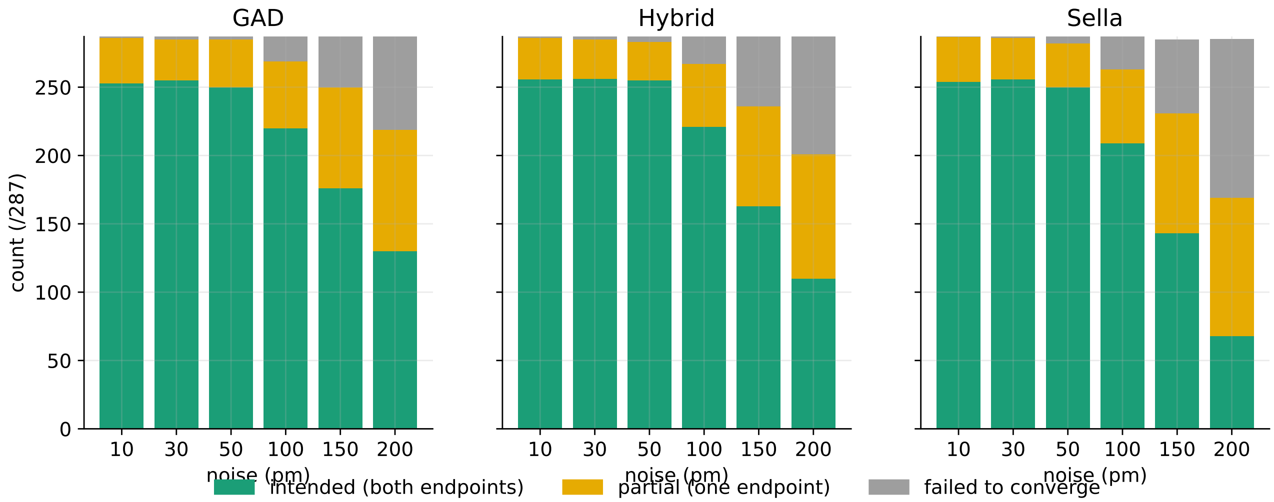


**Figure 4: Same breakdown, absolute counts (/287).** At 200 pm: intended/partial/failed = GAD 130/89/68 vs. Sella 68/101/116 — Sella's loss is non-convergence (116 vs. 68) plus one-sided partial saddles (101 vs. 89).

### 3.2 Mechanism: it is not wrong basins — it is one-sided saddles and non-convergence

- **Unintended ≈ 0.** The per-sample 5-tier IRC split gives unintended ≈ 0 for every method at every noise: 0 for GAD and the hybrid, and only 2 (150 pm) + 2 (200 pm) for Sella — **4 of 5166** samples total. *We therefore drop the earlier "Newton jumps across basin boundaries to unintended saddles" claim — it is unsupported by the data.*
- **What actually differs** (200 pm illustration; whole curves in Fig. 3): intended/partial/failed = **GAD 130/89/68** vs. **Sella 68/101/116**. Sella both fails to converge more *and* lands one-sided (partial) saddles more.
- **Why.** Small in-basin Euler steps preserve the link to *both* reactant and product, so even non-converged GAD geometries IRC-validate on both sides — intended can *exceed* convergence (150 pm: 61.3 vs. 55.1%; 200 pm: 45.3 vs. 40.8%). Larger Newton steps more often drift off one channel (→ partial) or fail to tighten (→ failed).
- **Convergence ≠ chemistry, stated correctly** (Fig. 5): it is *not* that convergence hides wrong saddles. (i) GAD recovers intended even when not strictly converged; (ii) Sella d=1 vs. d=3 — d=3 *converges more* but is *intended less* at every noise (a stale Hessian yields more partial/failed, not unintended).
- **Non-converged GAD breakdown** (why it fails to *converge*, not why it finds a wrong saddle): at 200 pm, of 163 non-converged, 71 (44%) drift to $n_{\text{neg}} \geq 2$ and 92 (56%) are plateau-orbits at $n_{\text{neg}}=1$ just above threshold (Section 3.3).

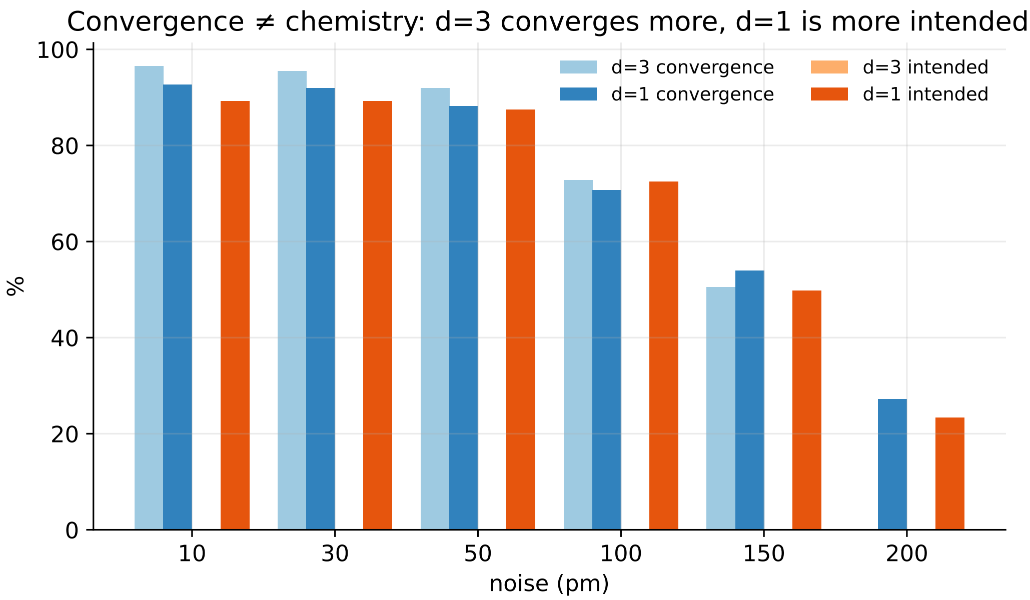


**Figure 5: Convergence ≠ chemistry.** Sella Hess.Freq.=3 converges more than Hess.Freq.=1 at low noise (a) yet recovers fewer intended reactions at every noise (b): the extra "convergences" are one-sided (partial) saddles, not unintended ones.

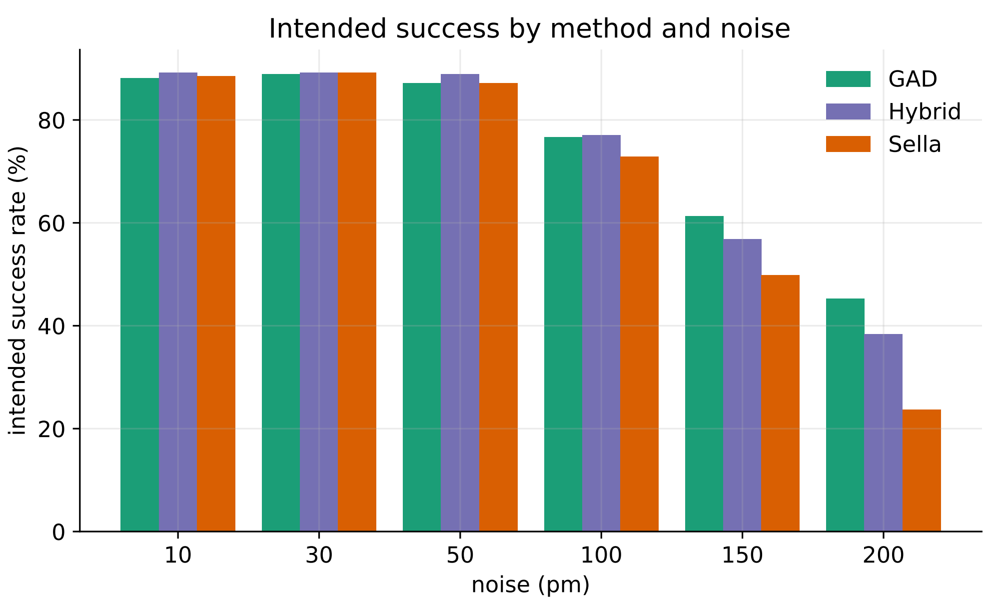


**Figure 6: Intended reactions recovered (counts/287).** The GAD family overtakes the baseline from 100 pm onward.

### 3.3 Secondary: saddle-convergence and the $F_{\max}$ plateau

Saddle-convergence ($n_{\text{neg}}{=}1 \wedge F_{\max}{<}\tau$) asks only whether the optimizer reached *a* first-order saddle, independent of which reaction it connects.

- **Crossover here too** (Fig. 7, Table 2). At low noise Sella's Newton step crushes the force fastest, so it leads saddle-convergence (92.7 vs. 89.2% @ 10 pm); GAD overtakes by 150 pm (55.1 vs. 54.0%)

and 200 pm (40.8 vs. 27.2%).

- **But convergence ≠ chemistry.** Sella's low-noise convergence lead does *not* convert into intended reactions (Table 1); the d=1/d=3 case makes this explicit (Fig. 5).

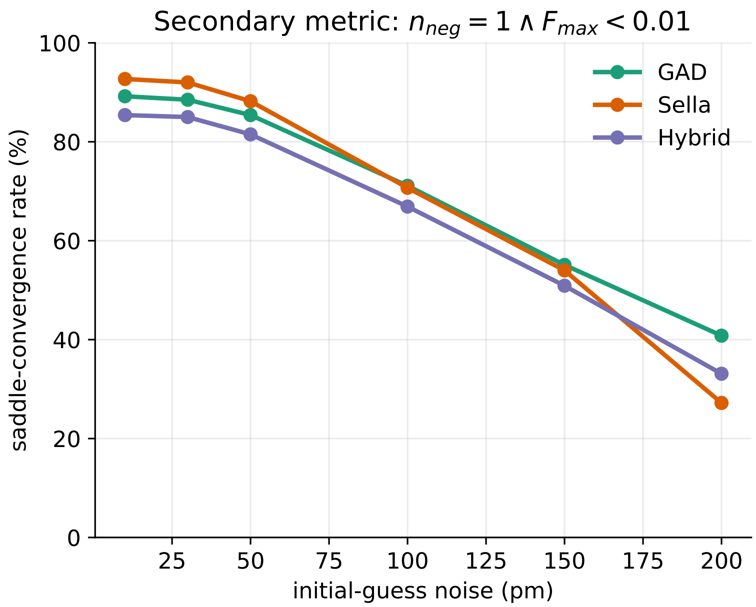


**Figure 7: Secondary metric: saddle-convergence rate** ($n_{\text{neg}}$=1 ∧ $F_{\max}$<0.01) vs. noise, $n = 287$. Sella leads at low noise; GAD overtakes at high noise — but convergence is not chemistry (cf. Fig. 1).

**Table 2:** Saddle-convergence rate (%, $n_{\text{neg}}$=1 ∧ $F_{\max}$<0.01, $n = 287$).

| **Saddle-convergence %** | **10 pm** | **30 pm** | **50 pm** | **100 pm** | **150 pm** | **200 pm** |
|---|---|---|---|---|---|---|
| GAD (ours) | 89.2 | 88.5 | 85.4 | 71.1 | **55.1** | **40.8** |
| Sella (baseline) | **92.7** | **92.0** | **88.2** | 70.7 | 54.0 | 27.2 |
| Hybrid (ours) | 85.4 | 85.0 | 81.5 | 66.9 | 50.9 | 33.1 |

**The $F_{\max}$ plateau.**

- **Plateau.** All GAD *dt* collapse to 0% at $F_{\max} < 0.005$ at every noise; Sella's Newton step retains 24–34% (low noise). Pure first-order dynamics has no samples in 0.001–0.005 eV/Å (Figs. 8, 9).

- **Mechanism.** GAD step $\sim dt\,\|\mathbf{F}\|$ shrinks linearly as $\mathbf{F} \to 0$ and falls below the eigenvector noise floor; Newton $\sim \|\mathbf{H}^{-1}\mathbf{F}\|$ stays finite. Structural to fixed-step first-order dynamics; larger *dt* trades it for instability.

- **Intrinsic.** 5× budget (10k vs. 2k steps, 50 pm) leaves $\tau$=0.01 unchanged (85.7%) and $\tau$<0.005 exactly 0.0%.

- **It lives on the secondary axis.** Intended success (Section 3.1) needs no sub-plateau force, so the plateau does not affect the primary metric.

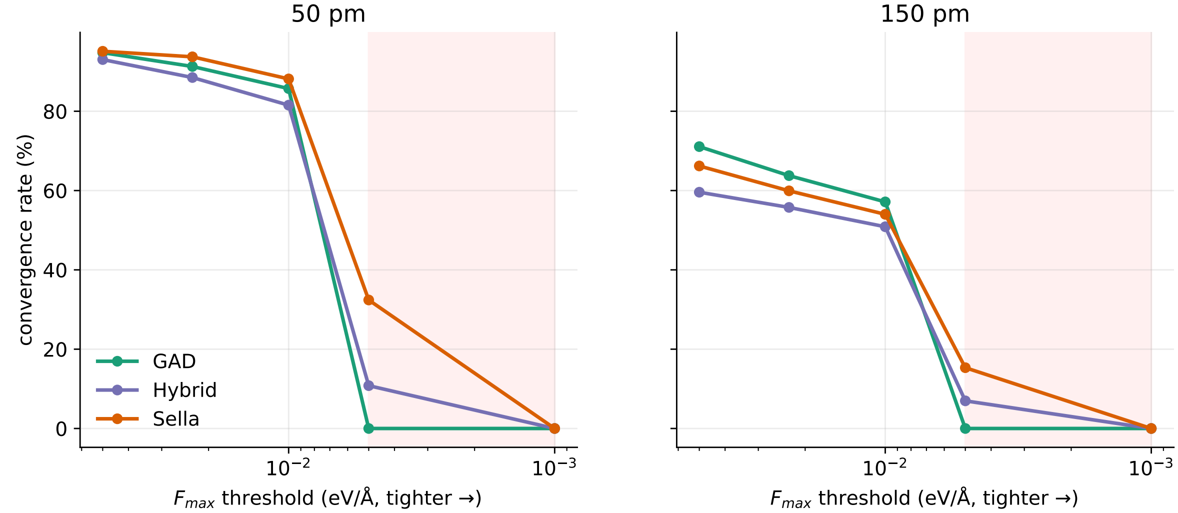


**Figure 8: The $F_{\max}$ plateau** (convergence vs. threshold; tighter to the right). GAD drops vertically near 0.01; Sella and the hybrid descend into 0.005 via Newton landing.

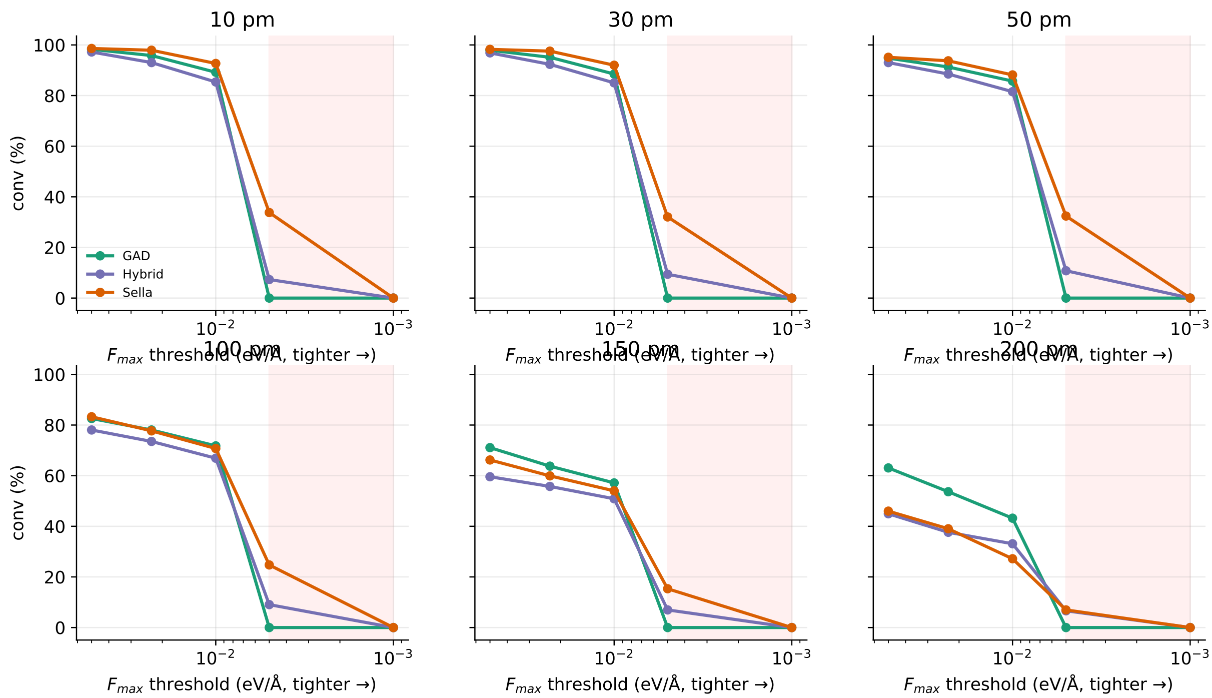


**Figure 9: Convergence vs. $F_{\max}$ threshold, all six noise levels.** The GAD cliff at 0.005 is universal.

### 3.4 Cost and geometry: the hybrid wins both

- **Wall-time / converged TS:** hybrid lowest at every noise — 1.3–3.0× vs. Sella, 3.6–4.5× vs. GAD (Fig. 10, Table 3). Few-but-expensive Sella steps (4–13) vs. many-cheap GAD steps (100–738) vs. intermediate-cheap hybrid (6–95) (Fig. 10). Pareto-dominant at high noise (Fig. 11).
- **Geometry:** converged-TS RMSD-to-reference — hybrid holds the tightest p95 at high noise (200 pm: **0.109** vs. GAD 0.456 vs. Sella 0.838 Å; 7.7× tighter than Sella). Sella's tail is bimodal

(Fig. 11, Table 3).

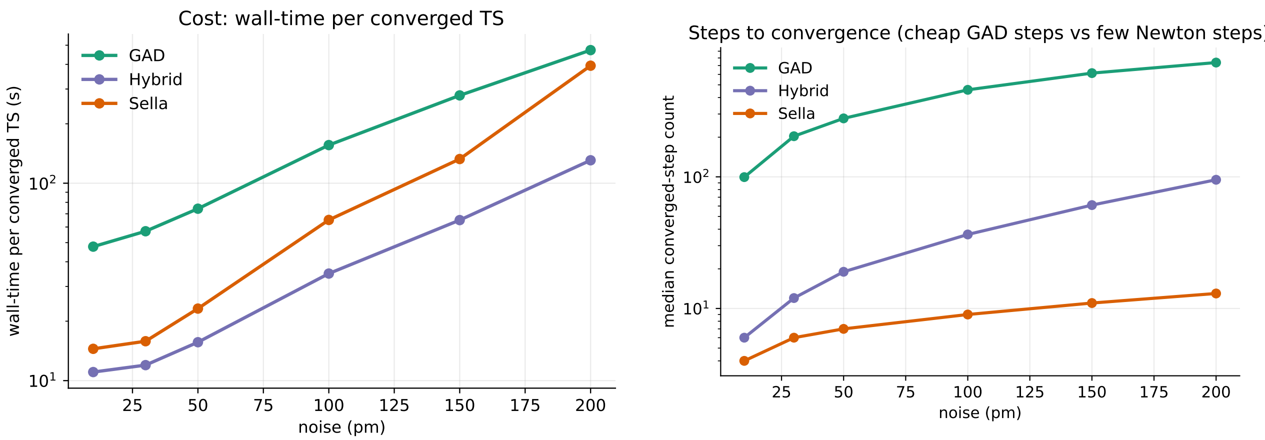


**Figure 10: Cost.** Wall-time per converged TS (left) and median steps (right). The hybrid is fastest despite more steps than Sella, because its steps are cheap.

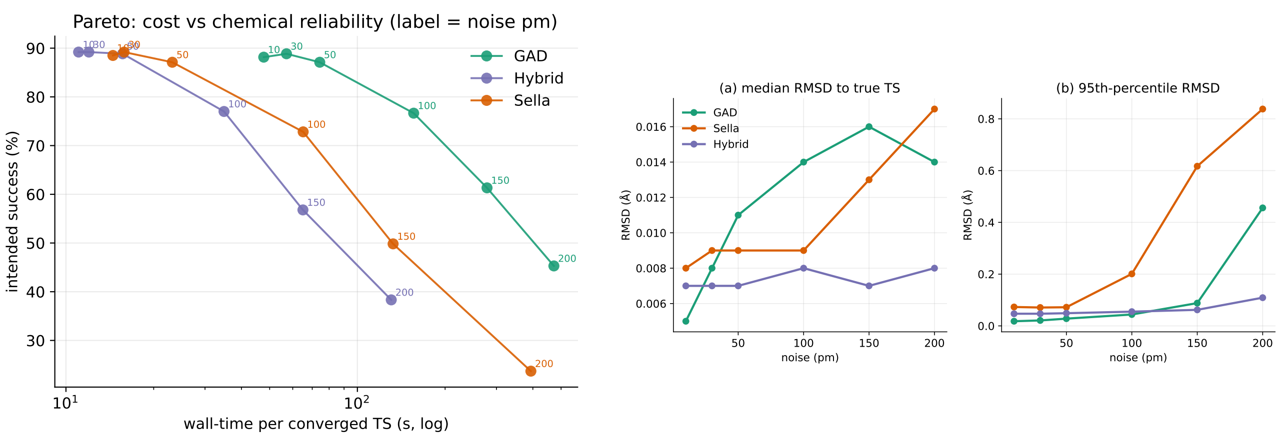


**Figure 11: Left: Pareto** (intended vs. wall-time; up-left is better) — the hybrid is dominant at high noise. **Right: RMSD to true TS** (median and p95) — the hybrid holds the tightest tail.

**Table 3:** Wall-time per converged TS (s) and converged-TS RMSD (Å, p95), $F_{\max} < 0.01$, $n = 287$.

| | | **10 pm** | **30 pm** | **50 pm** | **100 pm** | **150 pm** | **200 pm** |
|---|---|---|---|---|---|---|---|
| **wall/conv (s)** | **Hybrid** | **11.0** | **12.0** | **15.7** | **34.9** | **65.0** | **130.7** |
| | Sella | 14.5 | 15.8 | 23.2 | 65.2 | 132.6 | 393.8 |
| | GAD | 47.7 | 57.1 | 74.3 | 156.0 | 278.5 | 472.3 |
| **p95 RMSD (Å)** | **Hybrid** | 0.047 | 0.047 | 0.049 | **0.055** | **0.062** | **0.109** |
| | Sella | 0.073 | 0.071 | 0.072 | 0.201 | 0.617 | 0.838 |
| | GAD | 0.018 | 0.021 | 0.028 | 0.044 | 0.088 | 0.456 |

## 3.5 Scope boundary: GAD needs to start near the saddle

- Robustness above is for *perturbed-saddle* guesses. From a *reactant* start the GAD family is weak: hybrid ~**2%**, GAD $dt$=0.005 54% (partial $n$), while Sella reaches 81% (Fig. 12).

- Interpretation: far from the saddle the reaction mode is not yet identifiable as the lowest Eckart-projected eigenvector, so the GAD reflection has no correct direction to climb. GAD/hybrid are TS *refiners* given a near-saddle guess, not single-ended searchers from a minimum.

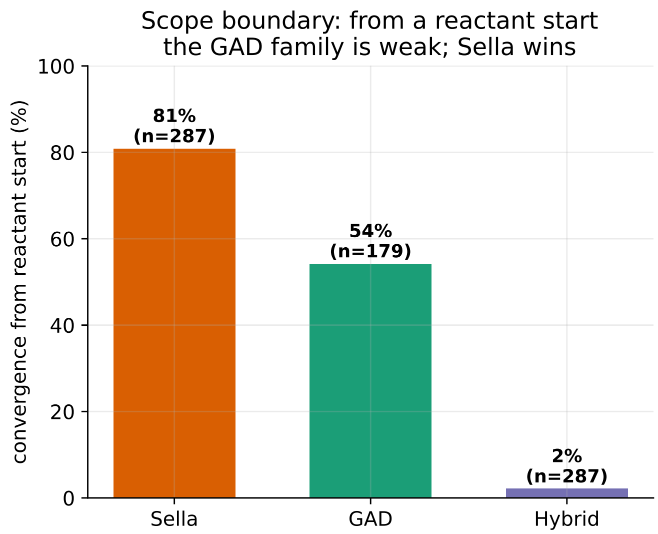


**Figure 12: Scope boundary.** Convergence from a reactant start: Sella wins; the GAD family needs a near-saddle guess.

# 4 Discussion

- When the analytic Hessian is cheap, the limiting factor is the step, not the curvature budget. A minimal first-order GAD walk is more robust on the metric that matters (intended success) precisely where large Newton/RFO jumps fail — under a degraded guess — while tying at low noise.
- The failure is *not* wrong basins: across all methods, converged geometries essentially never connect the wrong reaction. The high-noise gap is non-convergence plus one-sided (partial) saddles, which the small in-basin Euler step avoids.
- **Recommended default:** Hybrid GAD–Newton — fastest per converged TS, geometrically tightest, and chemically as reliable as GAD — for near-saddle (refinement) guesses.
- **Limitations:** (i) demonstrated for perturbed-saddle guesses; from a reactant start the GAD family is weak and Sella wins (Section 3.5). (ii) Convergence below $\tau$=0.001 is unreachable for all methods here. (iii) Isotropic perturbations; template/interpolation guesses carry structured error (future work).

## Appendix

# A Full tables (all nine configurations, $F_{\max} < 0.01$, $n = 287$)

**Table 4:** Intended-reaction success % (top) and saddle-convergence % (bottom), all nine configurations, from the aggregate IRC sweep. Bold = column best. The three headline methods (GAD, Sella cart+Eckart untuned d=1, Hybrid damped) were independently re-validated per-sample (`FINDING_IRC_5TIER_2026_06_23`; `irc_outcomes_5tier_test287.csv`); main-text numbers use that fresh re-validation and differ here by ≤2 pp, with unintended ≈ 0 confirmed.

| **Intended %** | **10 pm** | **30 pm** | **50 pm** | **100 pm** | **150 pm** | **200 pm** |
|---|---|---|---|---|---|---|
| GAD $dt = 0.003$ | 88.5 | **89.2** | **88.9** | **78.4** | 61.0 | **44.6** |
| GAD $dt = 0.005$ | 88.9 | **89.2** | **88.9** | 78.0 | **61.7** | **44.6** |
| GAD $dt = 0.007$ | 88.5 | 88.5 | 88.5 | **78.4** | **61.7** | 43.9 |
| Sella cart tuned d=1 | 88.9 | 88.9 | 87.5 | 70.7 | 46.7 | 17.8 |
| Sella cart+Eck untuned d=1 | **89.2** | **89.2** | 87.5 | 72.5 | 49.8 | 23.3 |
| Sella internal tuned d=1 | 88.2 | 87.5 | 83.3 | 64.5 | 33.1 | 16.0 |
| Sella cart+Eck untuned d=3 | 88.5 | 88.9 | 87.1 | 70.4 | 45.3 | 22.0 |
| Hybrid damped Eckart tr=0.05 | **89.2** | 88.9 | **88.9** | 76.7 | 57.5 | 38.7 |
| Hybrid undamped Eckart tr=0.05 | 88.9 | 88.9 | 88.5 | 77.0 | 57.1 | 38.7 |
| **Saddle-convergence %** | | | | | | |
| GAD $dt = 0.003$ | 89.2 | 88.5 | 85.4 | 71.1 | 55.1 | 40.8 |
| GAD $dt = 0.005$ | 89.2 | 88.5 | 85.7 | 71.8 | 57.1 | 43.2 |
| GAD $dt = 0.007$ | 89.2 | 88.9 | 85.7 | **72.8** | **58.2** | **44.6** |
| Sella cart tuned d=1 | 92.0 | 91.3 | 87.5 | 65.5 | 42.9 | 18.8 |
| Sella cart+Eck untuned d=1 | 92.7 | 92.0 | 88.2 | 70.7 | 54.0 | 27.2 |
| Sella internal tuned d=1 | 79.1 | 77.4 | 71.8 | 50.9 | 26.8 | 13.9 |
| Sella cart+Eck untuned d=3 | **96.5** | **95.5** | **92.0** | **72.8** | 50.5 | 23.3 |
| Hybrid damped Eckart tr=0.05 | 85.4 | 85.0 | 81.5 | 66.9 | 50.9 | 33.1 |
| Hybrid undamped Eckart tr=0.05 | 84.7 | 84.3 | 81.5 | 65.5 | 49.8 | 31.0 |

# B Comprehensive $F_{\max}$ sweep (convergence %, $n = 287$)

**Table 5:** Convergence at five thresholds; GAD block is 0.0% at $F_{\max} < 0.005$ everywhere.

| **thr** | **method** | **10 pm** | **30 pm** | **50 pm** | **100 pm** | **150 pm** | **200 pm** |
|---|---|---|---|---|---|---|---|
| 0.05 | GAD $dt$=0.005 | 98.3 | 97.9 | 94.8 | 82.6 | 71.1 | 63.1 |
| | Sella d=1 | 98.6 | 98.3 | 95.1 | 83.3 | 66.2 | 46.0 |
| | Hybrid damped | 97.2 | 96.9 | 93.0 | 78.0 | 59.6 | 44.9 |
| 0.01 | GAD $dt$=0.005 | 89.2 | 88.5 | 85.7 | 71.8 | 57.1 | 43.2 |
| | Sella d=1 | 92.7 | 92.0 | 88.2 | 70.7 | 54.0 | 27.2 |
| | Hybrid damped | 85.4 | 85.0 | 81.5 | 66.9 | 50.9 | 33.1 |
| 0.005 | GAD $dt$=0.005 | **0.0** | **0.0** | **0.0** | **0.0** | **0.0** | **0.0** |
| | Sella d=1 | 33.8 | 32.1 | 32.4 | 24.7 | 15.3 | 7.0 |
| | Hybrid damped | 7.3 | 9.4 | 10.8 | 9.1 | 7.0 | 6.6 |

# C Convergence-criterion family

Force thresholds (eV/Å): ASE default 0.05; Gaussian "Normal" max-force 0.0231 ($4.50 \times 10^{-4} E_h/a_0$); project canonical 0.01; tight 0.005; Sella documentation 0.001. $F_{\max}$ = maximum force component on any atom.

## D Sella baseline configurations

Three axes: **{cartesian|internal} {Eckart}? {tuned|untuned} Hess.Freq.=**$N$. Headline baseline: cartesian + Eckart, untuned, Hess.Freq.=1 (exact Hessian every step). Hess.Freq.=3 (library default) raises convergence at low noise (96.5 vs. 92.7% @ 10 pm) but lowers intended success at every noise (Fig. 5) — a stale Hessian yields more one-sided/failed outcomes, not unintended ones.

## E Reproducibility

- **Data (committed CSVs):** `master_2026_05_16.csv`, `noise_sweep_with_irc.csv`, `irc_topo_existing_methods.csv`, `irc_topo_hybrid.csv`, `threshold_sweep_2026_05_16.csv`, `rmsd_to_known_ts_compare.csv`, `reactant_0pm_2026_05_16.csv`, `test_summary_full.csv` (per-sample).
- **Figures:** `make_figs2.py`. **Code:** `core/gad.py` (Eq. 1), `projection/projection.py`, `search/hybrid_gad_eigfollownewton_eckart.py`, `search/irc_validate.py` (Sella IRC + VF2).
- Headline numbers are full $n = 287$. The 200 pm count illustration (130/89/68; 68/101/116) is from the latest local re-scoring and may differ by a few counts from the committed aggregate CSVs; McNemar significance is from the per-sample paired analysis.

## References


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